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# Efficacy of Tonlamarsen in Patients With Uncontrolled Hypertension

## The KARDINAL Phase 2 Randomized Clinical Trial

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## ABSTRACT

**BACKGROUND** Angiotensinogen production represents the rate-limiting step in activation of the renin-angiotensin-aldosterone system. Tonlamarsen is an investigational antisense oligonucleotide directed against hepatic angiotensinogen synthesis; its efficacy and safety among patients with hypertension are unknown.

**OBJECTIVES** The aim of this study was to assess the safety and efficacy of 90 mg tonlamarsen administered subcutaneously once monthly for 5 months compared with a single dose of tonlamarsen and subsequent placebo.

**METHODS** This randomized, placebo-controlled trial enrolled adults with office systolic blood pressure (BP) >135 mm Hg receiving 2 to 5 antihypertensive medications. Participants entered a 3-part treatment period with monthly administration of the study drug, consisting of a 4-week placebo lead-in, followed by 4-week active run-in with a single dose of tonlamarsen and subsequent randomization to 4 additional doses of tonlamarsen or matching placebo for 16 weeks. The coprimary endpoints were the between-group differences in the change from baseline to week 20 in plasma angiotensinogen and in office systolic BP.

**RESULTS** A total of 279 participants received placebo lead-in, 206 received 90 mg tonlamarsen during the active run-in, and 198 were randomized. The mean BP among randomized participants before and after the placebo lead-in was 147/90 and 147/89 mm Hg, respectively. Following active run-in with tonlamarsen, the mean BP was 140/87 mm Hg. Twenty weeks following the first dose of tonlamarsen, least squares (LS) mean percentage changes in plasma angiotensinogen levels were −23.0% (95% CI: −27.8% to −18.2%) with a single dose of tonlamarsen and subsequent placebo and −67.2% (95% CI: −71.9% to −62.4%) with monthly tonlamarsen administration, with a LS mean difference of −44.1% (97.5% CI: −51.9% to −36.4%; $P < 0.0001$). The LS mean changes in office systolic BP were −6.7 mm Hg (95% CI: −9.8 to −3.5 mm Hg) for participants following a single dose of tonlamarsen and subsequent placebo and −6.7 mm Hg (95% CI: −9.8 to −3.6 mm Hg) for participants treated with monthly tonlamarsen, with a LS mean difference of −0.1 mm Hg (95% CI: −4.5 to −4.4 mm Hg; $P = 0.97$). Serious adverse events were infrequent and similar between treatment groups.

**CONCLUSIONS** Among individuals with uncontrolled hypertension, monthly tonlamarsen administration was more effective at lowering plasma angiotensinogen compared with a single tonlamarsen dose, but there was no additional BP reduction. (A Study to Investigate Tonlamarsen for the Treatment of Adults With Uncontrolled Hypertension [KARDINAL]; NCT06864104) (JACC. 2026;■:■-■) © 2026 by the American College of Cardiology Foundation.

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**ABBREVIATIONS AND ACRONYMS**

**ACE** = angiotensin-converting enzyme

**AGT** = angiotensinogen

**ARB** = angiotensin receptor blocker

**BP** = blood pressure

**C5Research** = Cleveland Clinic Coordinating Center for Clinical Research

**RAAS** = renin-angiotensin-aldosterone system

Angiotensinogen (AGT), produced predominantly in the liver, is the rate-limiting substrate of the renin-angiotensin-aldosterone system (RAAS), and genetic variants that increase circulating AGT levels are associated with higher blood pressure (BP).[1,2] Globally, hypertension control rates are low, and one contributing factor is poor adherence to daily multipill regimens.[3,4] Lowering AGT with nucleic acid-based pharmacotherapies may result in stable and sustained BP lowering by targeting the most upstream component of the RAAS.[5,6]

Tonlamarsen is an investigational antisense oligonucleotide drug targeting *AGT* messenger RNA that is covalently bonded to triantennary n-acetyl galactosamine, a high-affinity ligand for the hepatocyte-specific asialoglycoprotein receptor. In small preliminary studies, weekly administration of an earlier version of a drug similar to tonlamarsen (IONIS-AGT-$L_{Rx}$) significantly lowered plasma AGT levels when given as monotherapy or when added to background antihypertensive medication regimens.[7] Compared with the prior drug, tonlamarsen has a shorter 16-nt sequence targeting the 3′ untranslated region of the *AGT* transcript and other features associated with enhanced hepatocyte targeting. These features may allow BP reduction with monthly or less frequent administration to improve patient adherence. However, the safety and efficacy of tonlamarsen are unknown. Prior studies of small interfering RNA drugs demonstrate suppression of hepatic AGT production and BP reduction among patients taking 0 or 1 background antihypertensive medication.[8,9] In the present trial, we examined the efficacy and safety of tonlamarsen in patients with uncontrolled hypertension taking 2 to 5 antihypertensive drugs.

## METHODS

**STUDY DESIGN, ORGANIZATION, AND OVERSIGHT.** KARDINAL (A Study to Investigate Tonlamarsen for the Treatment of Adults With Uncontrolled Hypertension) was a multicenter, prospective, randomized, active run-in, double-blind, placebo-controlled, phase 2 clinical trial performed at 39 sites within the United States. The trial was funded by Kardigan and designed by the sponsor and the Cleveland Clinic Coordinating Center for Clinical Research (C5Research). The trial was conducted in accordance with the principles of the Declaration of Helsinki and the Good Clinical Practice guidelines of the International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use. A central Institutional Review Board approved the study for each site (Advarra), and all participants provided written informed consent. An independent data monitoring committee provided patient safety oversight. Kardigan performed the statistical analyses, and C5Research independently confirmed the analyses.

**STUDY POPULATION.** The trial enrolled adults 18 years of age or older who were receiving stable doses of 2 to 5 antihypertensive medications with office systolic BP between 135 and 170 mm Hg. Exclusion criteria included known secondary causes of hypertension, an estimated glomerular filtration rate of <30 mL/min/1.73 m² body surface area, or serum potassium >5.1 mmol/L. Full inclusion and exclusion criteria are provided in the protocol in the Supplemental Appendix. Trial sites and participants were encouraged, but not mandated, to maintain the same background antihypertensive medications throughout the course of the trial.

**STUDY PROCEDURES.** Following screening, eligible participants entered a 3-part, 24-week treatment period, with subsequent 8-week safety follow-up (Supplemental Figure 1). Part A (week −4 to week 0) of the treatment period was a 1-time placebo subcutaneous injection. Four weeks later, eligible participants were administered a single 90-mg dose of tonlamarsen subcutaneously during an active run-in period (part B, weeks 0-4). In parts A and B, treatment was concealed from participants but not investigators. Part C (weeks 4-20) began 4 weeks after initial treatment with tonlamarsen, and participants were randomized 1:1 in a double-blind manner to 4 additional doses of tonlamarsen or matching placebo for the subsequent 16 weeks. Office BP was measured at weeks −4, 0, 4, 8, 12, 16, and 20. Baseline BP was measured 4 weeks after placebo administration, prior

The authors attest they are in compliance with human studies committees and animal welfare regulations of the authors' institutions and Food and Drug Administration guidelines, including patient consent where appropriate. For more information, visit the Author Center.

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to the first dose of tonlamarsen (week 0). Office BP measurement was performed using a sponsor-provided device equipped with an appropriately sized cuff (BP7255, Omron). After resting for 5 minutes, participants underwent a series of 3 BP measurements within a 10-minute period, with 2 to 3 minutes allowed between each reading. If the lowest and highest systolic BP measurements differed by >15 mm Hg, additional readings were performed. The last 3 consecutive, consistent systolic BP measurements were averaged to determine the final BP value.

Participants were instructed to perform daily home self-measurement of BP in the morning with an automated, brachial, oscillometric BP device (Blipcare BP800, Carematix). Participants were blinded to these results. Other biometrics, including heart rate, were also collected using sponsor-supplied wearable devices (Prolaio). Additional details describing the trial design and BP measurement are provided in the Supplemental Appendix.

**STUDY ENDPOINTS.** The coprimary endpoints were the percentage change from baseline (week 0) to week 20 in plasma AGT level and the change in office systolic BP among participants randomized to monthly tonlamarsen treatment, both compared with a single tonlamarsen dose and subsequent placebo. The key secondary endpoints were measured at week 20 and included the proportion of participants with office systolic BP <130 mm Hg, the change from baseline in mean self-assessed home systolic BP, the proportion of participants with any daily home systolic BP ≥150 mm Hg, and the change from baseline in mean office diastolic BP.

Safety endpoints included hyperkalemia (defined as a serum potassium >5.5 mmol/L), hypotension (systolic BP <90 mm Hg or between 90 and 100 mm Hg with symptoms or requiring clinical intervention), renal dysfunction (decrease of >30% from baseline in estimated glomerular filtration rate), and proteinuria (urine protein/creatinine ratio increase of >0.5 mg protein/mg creatinine from baseline). Prespecified adverse events of special interest included hypotension or hyperkalemia requiring clinical intervention, elevations in aminotransferases or bilirubin (aminotransferase measurements >3 times the upper limit of normal or total bilirubin >2 times the upper limit of normal), severe reductions in platelet count, defined as $<50{,}000/mm^3$ accompanied by bleeding, or any platelet measurement $<25{,}000/mm^3$.

**SAMPLE SIZE CALCULATION AND POWER.** The sample size estimate assumed 180 participants equally randomized to the 2 treatment groups: monthly placebo following a single dose of tonlamarsen or ongoing treatment with tonlamarsen 90 mg monthly for a total of 5 doses. The sample size calculation was based on a placebo-adjusted reduction of 5 mm Hg in office systolic BP with ongoing tonlamarsen between baseline and week 20, with a common SD of 10.2 mm Hg. This sample size provided 85% power at a 2-sided alpha level of 0.05.

**STATISTICAL ANALYSIS.** All randomized participants were analyzed using a mixed model for repeated measures to estimate the effect of monthly tonlamarsen compared with a single dose of tonlamarsen followed by placebo over time. The model included all available postbaseline response data. Fixed effects in the model included treatment group, race (Black vs non-Black), baseline value, visit, and treatment-by-visit interaction. The primary analysis was performed when all participants completed the week 20 visit or discontinued the study. Multiplicity testing for the coprimary endpoints used Holm's step-down procedure to control the overall family-wise error rate at $\alpha = 0.05$.[10] As specified by this procedure, the 2 *P* values were ordered from smallest to largest, and the smaller *P* value was first compared at $\alpha/2$ (0.025). If statistically significant, then the larger *P* value was compared at the full $\alpha$ level (0.05). For the secondary endpoints, a hierarchical testing approach gatekeeping procedure was used to control multiplicity (Supplemental Table 1).

## RESULTS

**STUDY PARTICIPANTS.** From February 2025 until August 2025, 519 patients were screened, 279 were administered placebo lead-in in part A, and 206 received active run-in with a single 90-mg dose of tonlamarsen in part B. An office systolic BP <135 mm Hg was the most common reason for participant ineligibility for part B, after receiving the placebo lead-in. Four weeks after the first tonlamarsen injection, 198 participants were randomized to either placebo (98 participants) or ongoing tonlamarsen 90 mg every 4 weeks for 16 weeks (100 participants) (Supplemental Figure 2). Baseline characteristics of randomized participants were similar between treatment groups (**Table 1**). The mean participant age was 61 years, 117 (59%) were men, 65 (33%) had diabetes, 21 (11%) had estimated glomerular filtration rates $<60$ mL/min/1.73 $m^2$, and 97 (49%) were Black. At trial enrollment, 118 participants (60%) were taking a 2-drug antihypertensive regimen, while others were taking 3 or more drugs. One hundred sixty-two participants (82%) were taking an angiotensin-converting enzyme (ACE)

**TABLE 1 Participants' Baseline Demographic and Clinical Characteristics**

| | Single Tonlamarsen Dose Followed by Placebo[a] (n = 98) | Tonlamarsen Administered Every 4 Weeks (n = 100) |
|---|---|---|
| Age, y | 61.0 ± 10.4 | 60.5 ± 10.0 |
| <65 | 59 (60.2) | 59 (59.0) |
| ≥65-<75 | 27 (27.6) | 35 (35.0) |
| ≥75 | 12 (12.2) | 6 (6.0) |
| Female | 39 (39.8) | 42 (42.0) |
| Race[b] | | |
| White | 47 (48.0) | 48 (48.0) |
| Black | 47 (48.0) | 50 (50.0) |
| Asian | 1 (1.0) | 2 (2.0) |
| Unspecified or other | 3 (3.1) | 0 (0.0) |
| Ethnicity | | |
| Hispanic or Latino | 17 (17.3) | 21 (21.0) |
| Body mass index, kg/m$^2$ | 32.0 ± 5.0 | 31.7 ± 5.4 |
| ≥18-<25 | 6 (6.1) | 11 (11.0) |
| ≥25-<30 | 23 (23.5) | 26 (26.0) |
| ≥30-<35 | 45 (45.9) | 36 (36.0) |
| ≥35 | 24 (24.5) | 27 (27.0) |
| Office blood pressure, mm Hg[c] | | |
| Systolic | 147.6 ± 9.4 | 145.9 ± 9.2 |
| Diastolic | 88.8 ± 10.6 | 89.4 ± 8.3 |
| Mean home blood pressure, mm Hg[d] | | |
| Systolic | 135.6 ± 9.3 | 134.5 ± 10.9 |
| Diastolic | 86.3 ± 10.1 | 87.2 ± 9.0 |
| Plasma angiotensinogen, mg/L | 23.7 ± 11.1 | 22.7 ± 9.3 |
| Estimated glomerular filtration rate,[e] mL/min/1.73 m$^2$ | 84.6 ± 18.1 | 84.8 ± 18.2 |
| <60 | 11 (11.2) | 10 (10.0) |
| ≥60 | 87 (88.8) | 90 (90.0) |
| Medical comorbidities | | |
| Diabetes | 30 (30.6) | 35 (35.0) |
| Hypercholesterolemia | 21 (21.4) | 25 (25.0) |
| Concomitant antihypertensive medications use | | |
| 2 medications | 57 (58.2) | 61 (61.0) |
| ≥3 medications | 41 (41.8) | 39 (39.0) |
| Use of ACE inhibitor or ARB | 82 (83.7) | 80 (80.0) |

Values are mean ± SD or n (%). Baseline characteristics are shown for all participants randomized into part C, the 16-week double-blind placebo-controlled treatment period. [a]Prior to randomization, participants assigned to placebo received a single injection of 90 mg tonlamarsen during part B, the 4-week active run-in period in which treatment was concealed only from the participant. [b]Race and ethnicity were self-reported by participants. [c]Office blood pressure obtained with the use of an automated oscillometric device during study visit at week 0 prior to tonlamarsen administration. [d]Home blood pressure was self-assessed through a study-provided automated oscillometric device whose results were concealed from study participants and electronically transmitted. This is the mean home blood pressure measurements obtained during part A, the 4-week placebo lead-in period in which treatment was concealed only from the participant. [e]Calculated using serum creatinine.
ACE = angiotensin-converting enzyme; ARB = angiotensin receptor blocker.

inhibitor or angiotensin receptor blocker (ARB) at the time of randomization.

Prior to the placebo lead-in period (part A), among randomized participants, the mean plasma AGT concentration was 22.1 mg/L, and mean office BP was 147/90 mm Hg. Four weeks later, prior to the start of the active run-in with tonlamarsen (part B), mean plasma AGT concentration and office BP were 23.2 mg/L and 147/89 mm Hg, respectively. Four weeks following the first dose of tonlamarsen, the mean plasma AGT concentration was 9.0 mg/L, and mean office BP was 140/87 mm Hg.

**PRIMARY ENDPOINT.** Results for the coprimary endpoints are reported in Table 2. At week 20, the least squares mean percentage change in plasma AGT from baseline (week 0) among participants treated with a single dose of tonlamarsen and then subsequently randomized to placebo for 16 weeks was −23.0% (95% CI: −27.8% to −18.2%). Among participants treated with 20 weeks of tonlamarsen, the change in plasma AGT was −67.2% (95% CI: −71.9% to −62.4%). The difference between treatment groups was −44.1% (97.5% CI: −51.9% to −36.4%; $P < 0.0001$). At week 20, the change in office systolic BP from baseline among participants treated with a single dose of tonlamarsen and subsequently randomized to placebo for 16 weeks was −6.7 mm Hg (95% CI: −9.8 to −3.5 mm Hg). Among participants treated with 20 weeks of tonlamarsen, the change in systolic BP was −6.7 mm Hg (95% CI: −9.8 to −3.6 mm Hg). The difference in systolic BP change between treatment groups was −0.1 mm Hg (95% CI: −4.5 to 4.4 mm Hg; $P = 0.97$). The trajectories of plasma AGT and office systolic BP changes throughout the trial are displayed in Figure 1. Results of the sensitivity analyses for the coprimary endpoints are displayed in Supplemental Table 2.

**SECONDARY ENDPOINTS.** Following 20 weeks of monthly tonlamarsen administration, participants did not achieve a systolic BP of <130 mm Hg more frequently than participants administered a single tonlamarsen dose with subsequent monthly placebo administration (26% vs 23%; OR: 1.1; 95% CI: 0.6-2.1) (Table 2). At week 20, the home systolic BP change from baseline among participants treated with a single dose of tonlamarsen and then randomized to placebo for 16 weeks was 2.8 mm Hg (95% CI: 0.4 to 5.2 mm Hg). Among participants assigned to 20 weeks of tonlamarsen, home systolic BP change was −0.3 mm Hg (95% CI: −2.8 to 2.2 mm Hg). Compared with a single dose of tonlamarsen and subsequent placebo, the home systolic BP change with ongoing tonlamarsen treatment was −3.1 mm Hg (95% CI: −6.6 to 0.4 mm Hg). There was no difference in office diastolic BP between treatment groups, and both treatment groups had similar proportions of

**TABLE 2 Tonlamarsen Primary and Key Secondary Endpoints**

| | Single Tonlamarsen Dose Followed by Placebo[a] (n = 98) | Tonlamarsen Administered Every 4 Weeks (n = 100) | Difference (95% CI) | OR (95% CI) | *P* Value[b] |
|---|---|---|---|---|---|
| Coprimary endpoints | | | | | |
| Percentage change from baseline (week 0) in plasma angiotensinogen at week 20[c] | −23.0 (−27.8 to −18.2) | −67.2 (−71.9 to −62.4) | −44.1 (97.5% CI: −51.9 to −36.4) | – | <0.0001 |
| Change from baseline in office systolic blood pressure at week 20,[c] mm Hg | −6.7 (−9.8 to −3.5) | −6.7 (−9.8 to −3.6) | −0.1 (−4.5 to 4.4) | – | 0.97 |
| Key secondary endpoints | | | | | |
| Change from baseline in mean self-assessed home systolic blood pressure at week 20,[c] mm Hg | 2.8 (0.4-5.2) | −0.3 (−2.8 to 2.2) | −3.1 (−6.6 to 0.4) | – | 0.08 |
| Change from baseline in office diastolic blood pressure at week 20,[c] mm Hg | −2.1 (−3.9 to −0.2) | −2.8 (−4.6 to −1.1) | −0.8 (−3.4 to 1.8) | – | 0.54 |
| Proportion of participants with office systolic blood pressure <130 mm Hg at week 20,[e] % | 23 (23.5) | 26 (26.0) | – | 1.07 (0.55-2.08) | 0.85 |
| Proportion of participants with any daily average home systolic blood pressure ≥150 mm Hg at week 20,[e] % | 31 (39.7) | 25 (37.3) | – | 0.90 (0.46-1.76) | 0.76 |

[a]Prior to randomization, participants assigned to placebo received a single injection of 90 mg tonlamarsen during part B, the 4-week active run-in period in which treatment was concealed only from the participant. [b]Between-group *P* values. Because one coprimary endpoint was not statistically significant, the secondary efficacy endpoints were analyzed for descriptive purposes only, and nominal *P* values were reported per the prespecified statistical analysis plan. [c]Least squares means, their 2-sided 95% CIs, and *P* values were estimated using a mixed model for repeated measures, which included all available postbaseline response data. The model had race (Black vs non-Black), treatment group, visit, and treatment-by-visit interaction as the fixed categorical effects and baseline value as the continuous covariates. [e]The ORs and their 2-sided 95% CIs were estimated using the logistic regression model, adjusted for race (Black vs non-Black), treatment group, and baseline office systolic blood pressure or baseline average home systolic blood pressure. Any missing week 20 data were imputed as nonresponse for office blood pressure. Only observed data were used for home blood pressure assessment.

daily home systolic BP averages of ≥150 mm Hg during week 20.

**EXPLORATORY ENDPOINTS.** Changes in systolic BP with ongoing monthly tonlamarsen treatment compared with a single dose and subsequent placebo among prespecified participant subgroups are shown in **Figure 2**. Supplemental Figure 3 displays prespecified BP changes among participants defined as early responders following active lead-in with tonlamarsen. Absolute changes in AGT through the course of trial are displayed in Supplemental Figure 4. Individual participant office systolic BP changes are reported in **Figure 3** and Supplemental Figure 5. The relationship between the change in AGT and office systolic BP change is displayed in Supplemental Figure 6.

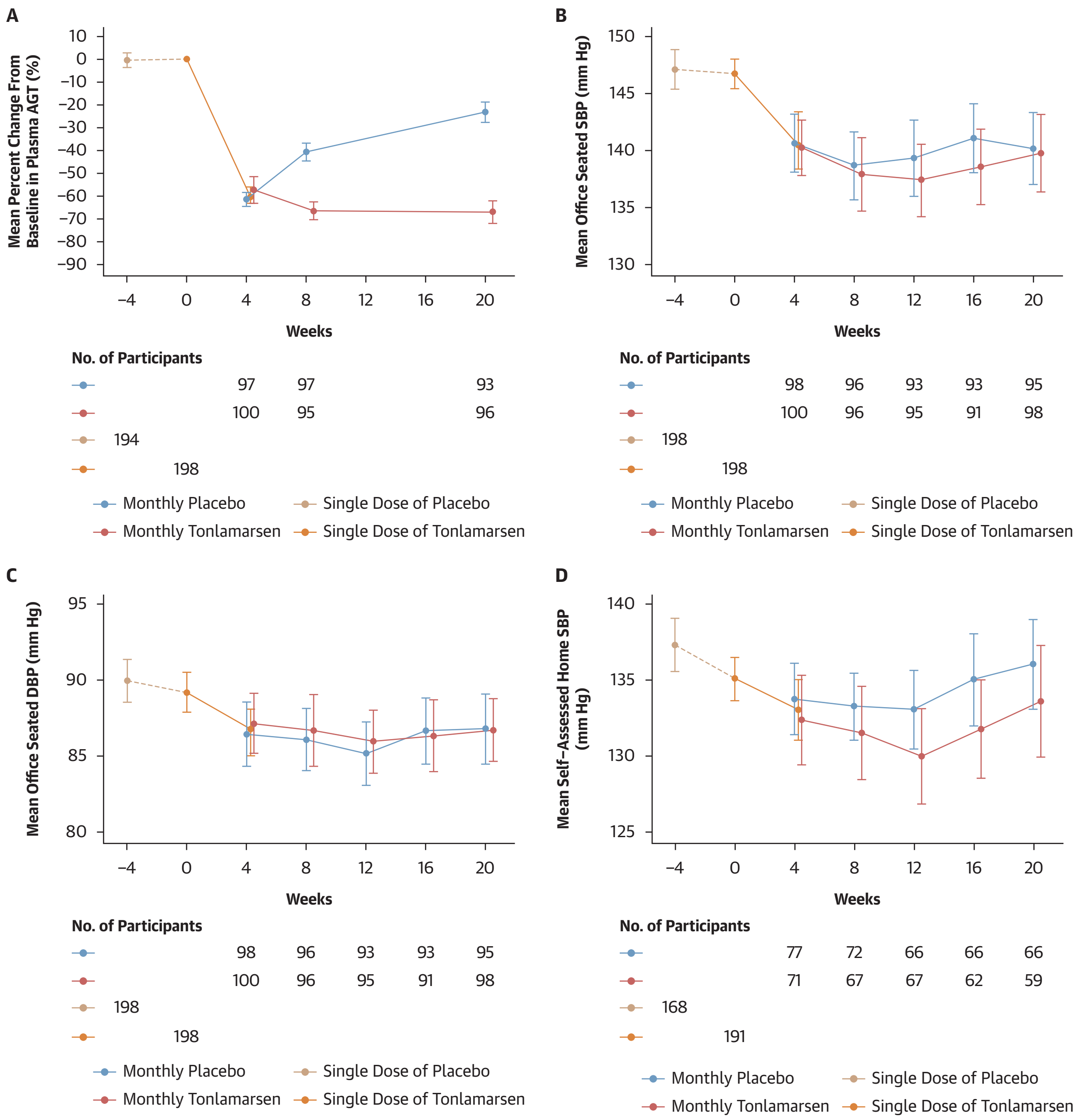
**SAFETY ENDPOINTS.** Intercurrent events for each treatment group are noted in Supplemental Table 3. Among participants who received a single dose of tonlamarsen and were subsequently randomized to placebo, serious adverse events occurred in 2 (2.1%). Among participants randomized to ongoing tonlamarsen treatment serious adverse events occurred in 5 (5.0%) (**Table 3**). A serum potassium concentration >5.5 mmol/L occurred in 2 participants randomized to placebo following a single dose of tonlamarsen and in no participants randomized to ongoing tonlamarsen treatment. Injection site reactions were more frequent with ongoing tonlamarsen treatment. Changes in selected safety biomarkers are shown in Supplemental Table 4. Rates of additional adverse events are shown in Supplemental Table 5.

One participant in the ongoing tonlamarsen treatment group died during the trial. The participant had hypertension, hyperlipidemia, obesity, and type 2 diabetes mellitus. He presented to the hospital 24 days after his last dose of study drug with respiratory distress and was found to have focal neurologic deficits and elevated BP. Computed tomographic angiography did not demonstrate any large vessel occlusion but did demonstrate an age-indeterminate right vertebral artery thrombus, and chest imaging revealed a right upper lung lobe infiltrate. Eight days following hospital admission, the patient died suddenly. The death was deemed unrelated to study drug by the site investigator.

## DISCUSSION

Among adults with uncontrolled hypertension, tonlamarsen 90 mg administered every 4 weeks for 20 weeks decreased plasma AGT by 67% from baseline compared with 23% among participants treated with a single dose of tonlamarsen and subsequent placebo. Although a reduction in systolic BP was observed during active run-in with tonlamarsen, ongoing monthly treatment did not result in more BP lowering than a single dose (**Central Illustration**). Injection site reactions were more common with

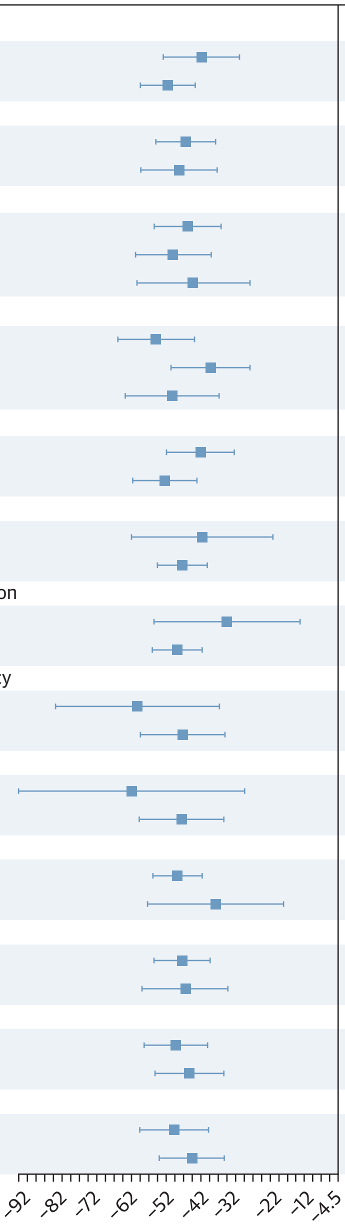
**FIGURE 1** Effect of Tonlamarsen on Plasma AGT and Blood Pressure




Shown are the trajectory of plasma angiotensinogen (AGT) (A), office systolic blood pressure (SBP) (B), office diastolic blood pressure (DBP) (C), and self-measured home SBP (D) among randomized participants. Part A (week −4 to week 0) was the placebo lead-in period, and treatment was concealed only from participants. Part B (weeks 0-4) was the tonlamarsen active run-in period, in which all trial participants received 90 mg tonlamarsen at week 0 and treatment was concealed only from participants. Part C began at the week 4 visit, and participants were randomized in a double-blind fashion to tonlamarsen every 4 weeks or matching placebo until week 20. Plasma AGT levels are shown as percentage change from baseline. The baseline measurement was defined as the measurement at week 0 (prior to tonlamarsen administration), and the coprimary endpoints were each assessed at week 20. The number of participants in each treatment arm with data available at each time point is shown below the x-axis. Data are presented as means and 95% CIs. Values are offset for readability.

**FIGURE 2** **Prespecified Subgroup Analyses for the Coprimary Endpoints**

**A** **Subgroup Analyses for Percent Change From Baseline in Plasma Angiotensinogen to Week 20**




| Subgroups | LS Mean Difference (95% CI) | Tonlamarsen /Tonlamarsen (N = 100) | Tonlamarsen /Placebo (N = 98) |
|---|---|---|---|
| Race | | | |
| Black | −39.17 (−50.06 to −28.27) | 50 (50.00) | 47 (47.96) |
| non-Black | −48.85 (−56.79 to −40.91) | 50 (50.00) | 51 (52.04) |
| Sex | | | |
| Male | −43.63 (−52.14 to −35.12) | 58 (58.00) | 59 (60.20) |
| Female | −45.61 (−56.52 to −34.69) | 42 (42.00) | 39 (39.80) |
| Age category | | | |
| <65 years | −43.10 (−52.65 to −33.55) | 59 (59.00) | 59 (60.20) |
| 65-75 years | −47.34 (−58.22 to −36.45) | 35 (35.00) | 27 (27.55) |
| ≥75 years | −41.52 (−57.76 to −25.28) | 6 (6.00) | 12 (12.24) |
| BMI category | | | |
| 18-30 kg/m$^2$ | −52.27 (−63.31 to −41.24) | 37 (37.00) | 29 (29.59) |
| 30-35 kg/m$^2$ | −36.53 (−47.76 to −25.30) | 36 (36.00) | 45 (45.92) |
| ≥35 kg/m$^2$ | −47.59 (−61.05 to −34.13) | 27 (27.00) | 24 (24.49) |
| Baseline AGT category | | | |
| >median | −39.43 (−49.15 to −29.70) | 47 (47.00) | 51 (52.04) |
| ≤median | −49.69 (−58.98 to −40.41) | 53 (53.00) | 47 (47.96) |
| History of hypertensive crisis | | | |
| Yes | −38.98 (−59.28 to −18.69) | 10 (10.00) | 8 (8.16) |
| No | −44.64 (−51.86 to −37.42) | 90 (90.00) | 90 (91.84) |
| History of hospitalization for hypertension | | | |
| Yes | −31.86 (−52.88 to −10.84) | 10 (10.00) | 14 (14.29) |
| No | −46.03 (−53.27 to −38.79) | 90 (90.00) | 84 (85.71) |
| History of hypertension during pregnancy | | | |
| Yes | −57.56 (−81.18 to −33.93) | 4 (9.52) | 5 (12.82) |
| No | −44.53 (−56.69 to −32.37) | 38 (90.48) | 34 (87.18) |
| History of pre-eclampsia or eclampsia | | | |
| Yes | −59.24 (−91.74 to −26.74) | 3 (7.14) | 5 (12.82) |
| No | −44.89 (−56.93 to −32.86) | 39 (92.86) | 34 (87.18) |
| Baseline ACEi/ARB medication used | | | |
| Yes | −46.15 (−53.28 to −39.02) | 81 (81.00) | 83 (84.69) |
| No | −35.02 (−54.58 to −15.47) | 19 (19.00) | 15 (15.31) |
| Baseline Diuretic BP medication used | | | |
| Yes | −44.76 (−52.78 to −36.74) | 57 (57.00) | 64 (65.31) |
| No | −43.80 (−56.17 to −31.43) | 43 (43.00) | 34 (34.69) |
| Early office SBP Responders * | | | |
| Yes | −46.54 (−55.70 to −37.37) | 47 (47.00) | 52 (53.61) |
| No | −42.59 (−52.53 to −32.65) | 53 (53.00) | 45 (46.39) |
| Baseline office SBP | | | |
| >median | −46.97 (−56.89 to −37.06) | 46 (46.00) | 53 (54.08) |
| ≤median | −41.92 (−51.29 to −32.55) | 54 (54.00) | 45 (45.92) |

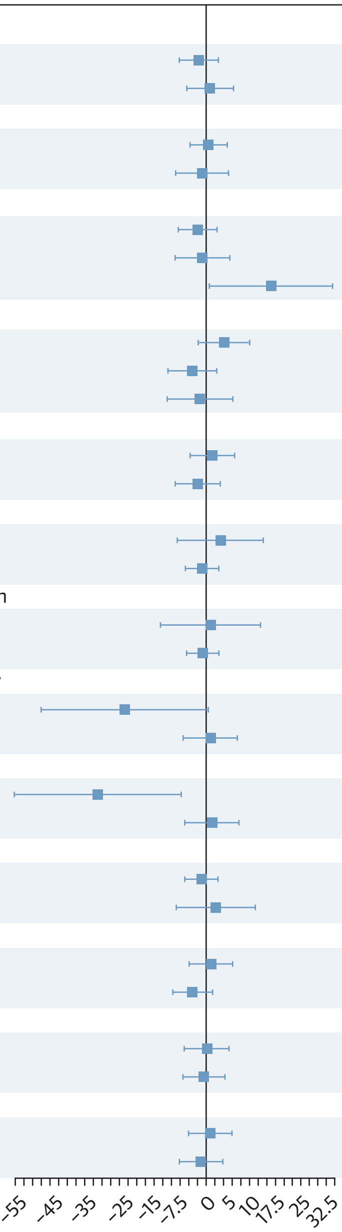
(A) Among prespecified participant subgroups, the difference in percentage change in AGT from baseline (week 0) to week 20 is shown among participants treated with a single dose of tonlamarsen and subsequently randomized to placebo compared with participants randomized to ongoing tonlamarsen treatment. (B) Difference in office systolic blood pressure change from baseline (week 0) to Week 20 between the 2 treatment groups. Error bars represent 95% CIs. *Early responders were defined as randomized participants achieving decreases in office SBP from baseline (week 0) to week 4 greater than or equal to the median decrease (−5.5 mm Hg) among all participants with data available at both visits. ACEi = angiotensin-converting enzyme inhibitor; ARB = angiotensin receptor blocker; BMI = body mass index; BP = blood pressure; LS = least squares; other abbreviations as in Figure 1.

*Continued on the next page*

**FIGURE 2** Continued

**B** Subgroup Analyses for Change From Baseline in Mean Office Seated SBP to Week 20

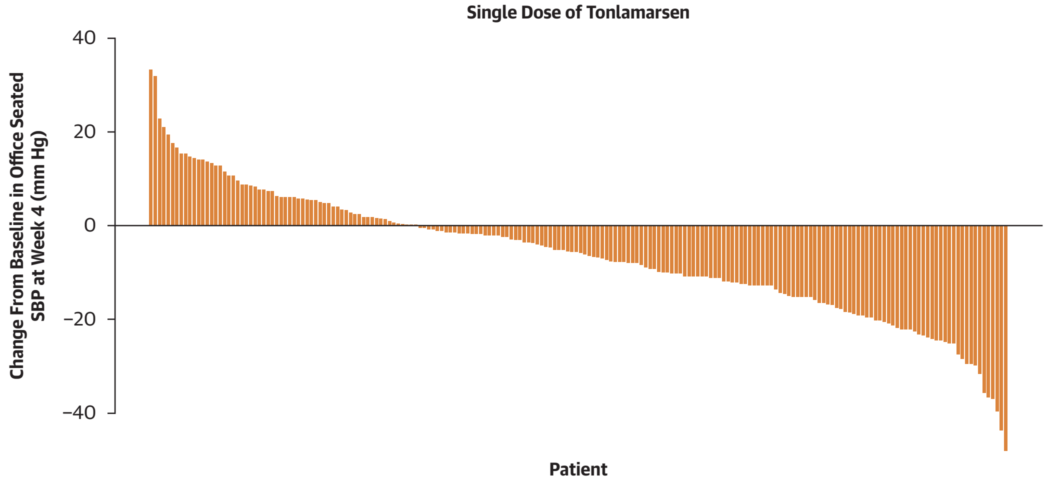
| Subgroups | LS Mean Difference (95% CI) | Tonlamarsen /Tonlamarsen (N = 100) | Tonlamarsen /Placebo (N = 98) |
|---|---|---|---|
| Race | | | |
| Black | −1.76 (−7.50 to 3.97) | 50 (50.00) | 47 (47.96) |
| non-Black | 1.44 (−5.26 to 8.15) | 50 (50.00) | 51 (52.04) |
| Sex | | | |
| Male | 0.99 (−4.40 to 6.37) | 58 (58.00) | 59 (60.20) |
| Female | −0.89 (−8.55 to 6.77) | 42 (42.00) | 39 (39.80) |
| Age category | | | |
| <65 years | −2.11 (−7.69 to 3.47) | 59 (59.00) | 59 (60.20) |
| 65-75 years | −0.72 (−8.51 to 7.08) | 35 (35.00) | 27 (27.55) |
| ≥75 years | 19.05 (1.33-36.77) | 6 (6.00) | 12 (12.24) |
| BMI category | | | |
| 18-30 kg/m$^2$ | 5.42 (−1.98 to 12.82) | 37 (37.00) | 29 (29.59) |
| 30-35 kg/m$^2$ | −3.69 (−10.64 to 3.25) | 36 (36.00) | 45 (45.92) |
| ≥35 kg/m$^2$ | −1.50 (−11.01 to 8.02) | 27 (27.00) | 24 (24.49) |
| Baseline AGT category | | | |
| >median | 2.08 (−4.16 to 8.31) | 47 (47.00) | 51 (52.04) |
| ≤median | −2.13 (−8.54 to 4.28) | 53 (53.00) | 47 (47.96) |
| History of hypertensive crisis | | | |
| Yes | 4.48 (−7.90 to 16.85) | 10 (10.00) | 8 (8.16) |
| No | −0.73 (−5.52 to 4.06) | 90 (90.00) | 90 (91.84) |
| History of hospitalization for hypertension | | | |
| Yes | 1.52 (−12.94 to 15.97) | 10 (10.00) | 14 (14.29) |
| No | −0.55 (−5.25 to 4.15) | 90 (90.00) | 84 (85.71) |
| History of hypertension during pregnancy | | | |
| Yes | −23.15 (−47.32 to 1.03) | 4 (9.52) | 5 (12.82) |
| No | 1.53 (−6.27 to 9.33) | 38 (90.48) | 34 (87.18) |
| History of pre-eclampsia or eclampsia | | | |
| Yes | −30.92 (−54.96 to −6.89) | 3 (7.14) | 5 (12.82) |
| No | 1.98 (−5.78 to 9.75) | 39 (92.86) | 34 (87.18) |
| Baseline ACEi/ARB medication used | | | |
| Yes | −1.12 (−5.92 to 3.68) | 81 (81.00) | 83 (84.69) |
| No | 3.11 (−8.25 to 14.46) | 19 (19.00) | 15 (15.31) |
| Baseline Diuretic BP medication used | | | |
| Yes | 1.70 (−4.58 to 7.99) | 57 (57.00) | 64 (65.31) |
| No | −3.67 (−9.43 to 2.08) | 43 (43.00) | 34 (34.69) |
| Early office SBP Responders * | | | |
| Yes | 0.40 (−6.04 to 6.84) | 47 (47.00) | 52 (53.61) |
| No | −0.35 (−6.36 to 5.66) | 53 (53.00) | 45 (46.39) |
| Baseline office SBP | | | |
| >median | 1.33 (−4.74 to 7.39) | 46 (46.00) | 53 (54.08) |
| ≤median | −1.23 (−7.57 to 5.10) | 54 (54.00) | 45 (45.92) |




multiple doses of tonlamarsen, but other adverse event rates were similar among participants randomized to placebo following a single dose of tonlamarsen.

Nucleic acid-based therapeutics decreasing hepatic AGT synthesis represent a new approach to RAAS modulation with drugs that are durable and not dependent on daily adherence. Hepatic AGT suppression targets the dominant source of circulating AGT and may provide more sustained reduction across dosing intervals in the downstream products of AGT, such as angiotensin II. However, it remains

**FIGURE 3** Individual Participant Office Systolic Blood Pressure Changes 4 Weeks After Initial Tonlamarsen Administration




Waterfall plot showing the change from baseline in office systolic blood pressure (SBP) following a single dose of tonlamarsen during the active run-in period, part B (weeks 0-4). Each vertical bar represents an individual participant, and bars are ordered from the greatest increase to the greatest decrease in SBP. Negative values indicate reductions in SBP relative to baseline.

uncertain whether reduction of hepatic AGT synthesis results in BP reduction among patients already treated with 2 or more antihypertensive medications that may include an ACE inhibitor or ARB.

Interpretation of the results of the KARDINAL trial is complicated by the unanticipated persistence of AGT suppression among participants administered a single tonlamarsen dose followed by monthly placebo. Tonlamarsen has an elimination half-life of between 2 and 4 weeks, which would be expected to result in limited to no circulating drug between 10 and 20 weeks following a single dose. However, mean plasma AGT levels were still suppressed by 23% compared with baseline even 20 weeks after one-time drug exposure.

Given these findings, potential explanations for the AGT and BP changes attributed to tonlamarsen during the course of the trial include the following: decreasing hepatic production of AGT may not reduce BP among patients receiving multiple antihypertensive medications, residual AGT suppression observed at week 20 among placebo-assigned participants may have resulted in greater BP reduction than expected, or tonlamarsen may lower office systolic BP between 6 and 7 mm Hg within 4 weeks of an initial dose, and this reduction is maintained over 20 weeks even as AGT levels rise.

When zilebesiran, a small interfering RNA that also suppresses hepatic AGT production, was administered to patients not taking background antihypertensive therapy in early trials, lower residual AGT was associated with greater BP lowering.[8] In contrast, among patients treated with only with the ARB olmesartan, BP lowering attributed to zilebesiran was attenuated.[9] The ideal amount of AGT reduction among patients being treated with multiple antihypertensive medications is unknown. One mechanistic factor that may affect the relationship between AGT lowering and BP is the role of angiotensin II, the principal downstream product of AGT, which is classically viewed as a vasoconstrictor through angiotensin II type 1 receptor activation but can also exert vasodilatory effects via angiotensin II type 2 receptors. AGT also serves as the precursor for multiple biologically active peptides, including angiotensin-(1-7), which has vasodilatory properties.[11,12] Among patients taking ACE inhibitors or ARBs, reductions in hepatic AGT production may produce smaller incremental effects on angiotensin II generation and vascular tone, thereby blunting the observed BP response. However, if drugs that decrease hepatic AGT synthesis do not meaningfully reduce BP when taken in combination with multiple antihypertensive medications, the BP reduction observed during the active tonlamarsen run-in period of this trial would have to be attributed to additional regression to the mean in office BP measurements. This effect would have had to occur

**TABLE 3 Adverse Events Occurring in the Primary Safety Population Period**

| | Single Tonlamarsen Dose Followed by Placebo[a] (n = 97) | Tonlamarsen Administered Every 4 Weeks (n = 100) |
|---|---|---|
| Participants with any treatment emergent adverse event[b] | | |
| Any grade | 36 (37) | 46 (46) |
| Grade 1 | 18 (19) | 24 (24) |
| Grade 2 | 15 (16) | 16 (16) |
| Grade 3 | 3 (3) | 5 (5) |
| Grade 4 | 0 | 0 |
| Grade 5 | 0 | 1 (1) |
| Participants with any serious treatment emergent adverse event[c] | 2 (2) | 5 (5) |
| Death[d] | 0 | 1 (1) |
| Serum potassium >5.5 mmol/L | 2 (2) | 0 |
| Decrease of >30% from baseline in estimated glomerular filtration rate | 3 (3) | 3 (3) |
| Increase in urine protein/creatinine ratio of >0.5 mg protein/mg creatinine from baseline | 2 (2) | 2 (2) |
| Injection site reaction | 3 (3) | 19 (19) |
| Treatment emergent adverse events of special interest | | |
| Participants with any treatment emergent adverse event of special interest | 1 (1) | 2 (2) |
| Hypotension[e] | 0 | 1 (1) |
| Hyperkalemia requiring clinical intervention | 1 (1) | 0 |
| Elevations in aminotransferases[f] | 0 | 1 (1) |
| Severe reduction in platelet count[g] | 0 | 0 |

Values are n (%). The primary safety population is all participants who received one dose of study drug following randomization. One participant received a single dose of tonlamarsen during the active run-in (part B) and was subsequently randomized to placebo but was never administered any placebo doses following randomization. [a]Prior to randomization, participants assigned to placebo received a single injection of 90 mg tonlamarsen during part B, the 4-week active run-in period in which treatment was concealed only from the participant. [b]Graded using the Common Terminology Criteria for Adverse Events, where grade 1 is mild, grade 2 is moderate, grade 3 is severe or disabling, grade 4 is life threatening, and grade 5 is death. [c]In the single dose of tonlamarsen with subsequent placebo arm, 2 participants experienced serious adverse events: 1 instance of increased blood glucose and 1 skull fracture with subarachnoid hematoma. In the monthly tonlamarsen arm, 5 participants experienced serious adverse events, including angioedema, unstable angina, sudden cardiac death, and loss of consciousness. Additionally, 1 participant experienced sepsis, acute kidney injury, and acute myocardial infarction. [d]A single instance of sudden cardiac death was reported among the participants receiving monthly tonlamarsen injections; the patient had been admitted to the hospital for respiratory distress and stroke-like symptoms. [e]Systolic blood pressure <90 mm Hg or between 90 and 100 mmHg with symptoms or requiring clinical intervention. [f]Aminotransferase measurement >3 times the upper limit of normal or total bilirubin >2 times the upper limit of normal. [g]Measurement <50,000/mm3 accompanied by bleeding or any measurement <25,000/mm$^3$.

despite a 4-week placebo lead-in and would have had to be maintained throughout the subsequent 16 weeks of follow-up.

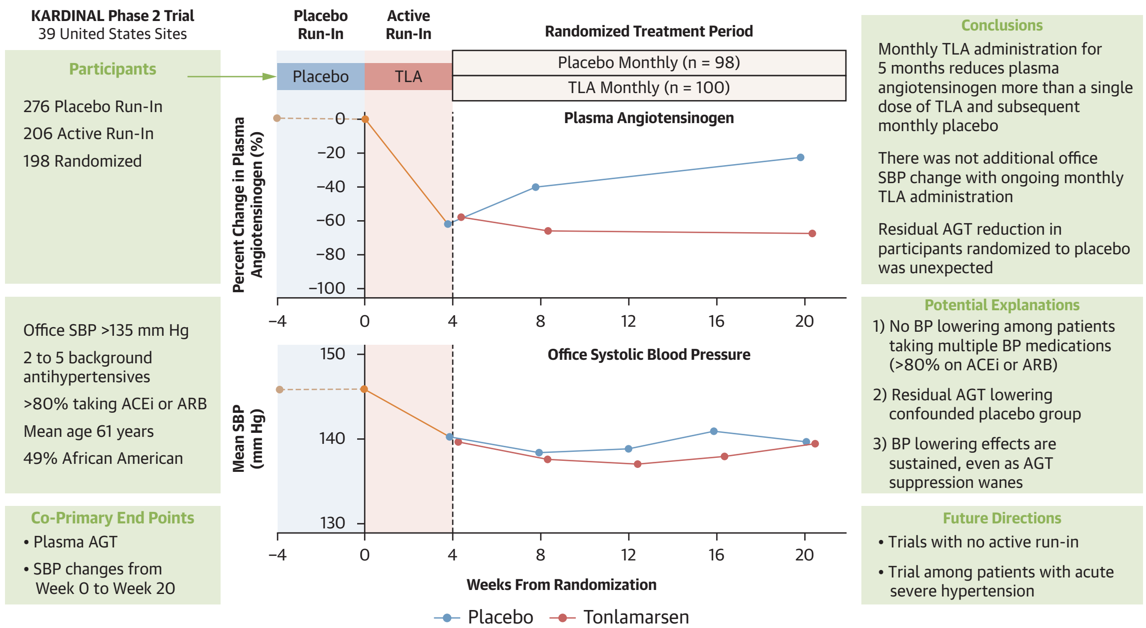
A second possible explanation is that persistent AGT suppression among participants assigned to placebo following a single tonlamarsen dose 20 weeks earlier produced greater BP reduction than would be expected with placebo alone in the absence of a prior tonlamarsen exposure. This carryover effect may have violated the trial's prespecified statistical assumptions, and future studies should include a placebo group without a tonlamarsen run-in.

A third potential explanation is that a single dose of tonlamarsen may produce a reduction in office systolic BP within 4 weeks that is sustained for at least 20 weeks, even as circulating AGT levels gradually increase, as observed among participants randomized to placebo. Continuous suppression of AGT may not be required to achieve sustained downstream physiological effects, despite partial recovery of the upstream substrate, possibly because of RAAS hysteresis as well as renal and vascular adaptation. Additionally, short-term interruption of RAAS activity may attenuate neurohormonal feedback mechanisms that otherwise sustain hypertension. Renal, vascular, and neurohormonal alterations following initial hepatic AGT suppression may maintain BP reductions even as AGT concentrations rise toward baseline. If confirmed in subsequent studies, this pattern would suggest that sustained physiological benefit may not require continuous AGT suppression and would have implications for future dosing strategies and trial design.

It remains to be determined whether specific groups of patients with uncontrolled hypertension may or may not benefit from drugs that suppress AGT production, particularly given that many patients achieve adequate BP control if prescribed guideline-recommended combination therapy and remain adherent.[13,14] One population that may benefit from an infrequently administered, long-acting therapy is patients with poor adherence to daily antihypertensive medication regimens. Nonadherence is common, is especially prevalent among patients with apparent treatment-resistant hypertension with estimates varying between 20% and 46%, and is associated with higher risks for all-cause and cardiovascular mortality, as well as worse clinical outcomes.[15,16] However, among currently available classes of BP-lowering medications, ARBs and ACE inhibitors demonstrate the highest rates of adherence.[17]

Beyond addressing nonadherence, certain patient populations may be particularly well suited for therapies that suppress hepatic AGT production, particularly those in whom genetic and physiological factors are associated with increased AGT expression or heightened RAAS activity. For example, women with a history of hypertensive disorder of pregnancy have evidence of heightened angiotensin II sensitivity, there are genetic associations between AGT variants and preeclampsia, and preclinical models demonstrate BP improvement with hepatic AGT silencing.[18-20] It is possible that directly reducing hepatic AGT production could be an effective strategy in this high-risk population, in which long-term

**CENTRAL ILLUSTRATION** The KARDINAL Phase 2 Randomized Controlled Trial of Tonlamarsen




**Laffin LJ, et al. JACC. 2026;■(■):■-■.**

ACEi = angiotensin-converting enzyme inhibitor; AGT = angiotensinogen; ARB = angiotensin receptor blocker; BP = blood pressure; KARDINAL = A Study to Investigate Tonlamarsen for the Treatment of Adults With Uncontrolled Hypertension; SBP = systolic blood pressure; TLA = tonlamarsen.

cardiovascular risk remains elevated after pregnancy. The total number of participants with a history of a hypertensive disorder of pregnancy enrolled in the current trial was limited. Efficacy assessment in this patient population will require study in larger, dedicated trials.

Patients presenting with acute severe hypertension represent another potential target group, as these episodes are similarly characterized by marked RAAS activation and may be incompletely controlled with conventional RAAS blockade because of compensatory increases in renin and continued substrate availability.[21] A single or limited number of doses of a drug leading to hepatic AGT suppression may facilitate transition to a lower postdischarge BP setpoint. Whether such an AGT suppression strategy can improve BP stability and reduce the risk for recurrent acute severe hypertension will require dedicated trials.

**STUDY LIMITATIONS.** First, the prolonged suppression of plasma AGT seen with a single 90-mg dose of tonlamarsen was unexpected. Although there was a significant difference in percent AGT lowering between treatment groups 20 weeks following a single dose of tonlamarsen, there was still a mean reduction of 23% from baseline in serum AGT levels. Subsequent trials will need to compare tonlamarsen strictly with placebo and forgo an active run-in period to definitively demonstrate the magnitude of BP reduction.

Second, the trial was conducted exclusively at sites within the United States, and the generalizability of these findings to patients in other countries remains to be established.

Third, the number of participants and length of the trial limited robust assessment of the impact of tonlamarsen on cardiovascular outcomes.

## CONCLUSIONS

Treatment with 90 mg tonlamarsen monthly for a total of 5 months among patients with uncontrolled hypertension taking 2 or more antihypertensive medications, with >80% receiving ACE inhibitors or ARB, lowered plasma AGT more than a single dose of tonlamarsen and subsequent placebo. However, similar office systolic BP lowering was observed between treatment groups at week 20 despite partial recovery of plasma AGT. The prolonged and unanticipated AGT reduction among participants who received a single dose of tonlamarsen necessitates additional placebo-controlled trials assessing single or multidose tonlamarsen regimens.

**DATA AVAILABILITY** The data from the KARDINAL trial will not be shared.

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**KEY WORDS** angiotensinogen, clinical trial, hypertension

**APPENDIX** For KARDINAL study sites and investigators, KARDINAL trial leadership, inclusion and exclusion criteria, BP measurement, and supplemental figures and tables, please see the online version of this paper.